

Cyrus Guided Notes: MIT Missing Semester

MIT course pack

2026-05-15

Course source

Official source: <https://missing.csail.mit.edu/>

Why this course is in the system. Engineering tools course for Git, shell, debugging, data processing, and learning-system maintenance.

Learning output. Use Notion as the tool readiness checklist so environment blockers are visible.

Prerequisite checkpoint

- Terminal basics
- Files and paths
- Git basics

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned}\text{pipeline} &= f_n \circ f_{n-1} \circ \dots \circ f_1 \\ \text{commit} &= H(\text{tree}, \text{parents}, \text{message}) \\ \text{debugging} &= \text{reproduce} \rightarrow \text{isolate} \rightarrow \text{verify} \\ \text{automation} &= \text{input} \rightarrow \text{script} \rightarrow \text{checked output}\end{aligned}$$

Visual page

Draw shell, editor, git, tests, and browser as one reproducible learning loop.

GoodNotes pages

- Page MIT-T001: shell and git workflow
- Page MIT-T002: debugging checklist

Web interaction

A command checklist that marks environment readiness.

Obsidian and Notion

Layer	Output
Obsidian	MIT Tools -> Missing Semester
Notion	Tool blocker, command evidence, and next action.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.